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Brain Structure Volumes in the Mole Rat, *Spalax ehrenbergi* (Spalacidae, Rodentia) in Comparison to the Rat and Subterrestrial Insectivores

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With 3 Figures and 5 Tables

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Summary: Natural blindness and a subterranean, digging mode of life demand peculiar adaptations of the central nervous system in the mole rat *Spalax ehrenbergi*, which are the focus of this quantitative investigation.

Volumes of 25 brain structures in *Spalax* were evaluated allometrically, using the least encephalized mammalian species, the Madagassian hedgehog-like tenrecs (Tenrecinae) as a reference base, and their sizes compared with those of the rat (as a more generalized representative of rodents) and of some subterranean Insectivora.

The allometric approach reveals that *Spalax* has a larger brain than tenrecs and the rat. Within the brain, the neocortex and diencephalon are well developed, an observation also made in other mammalian species with a relatively high encephalization.

An unique feature in *Spalax* is the enlargement of motor structures of the brain, such as the cerebellum (and cerebellar nuclei), and the striatum. Most conspicuous is the large size of the nucleus motorius nervi trigemini, reflecting the importance of masticatory muscles for the special digging technique, which demand an intense use of the teeth for loosening the soil.

Key words: brain size, allometry, sensory nuclei, motor nuclei

Introduction

The mole rat *Spalax* is a subterranean rodent, which is distributed in the eastern Mediterranean, eastern Europe, Asia Minor, and southern Russia.

Throughout the last few decades, interest has been focussed on the process of active speciation in the *Spalax ehrenbergi* superspecies in Israel (for review see Nevo, 1991).

Of special interest is the composition of the brain in such a naturally blind animal, and several attempts have been made to answer the questions:

a) What happens to those same brain areas which control visual functions in sighted animals (Leder, 1974, 1975; Bronchti et al., 1989, 1991; Necker et al., 1992; Rehkämper et al., 1994)?

b) Is there a compensation in other sensory modalities for the loss of vision (Leder, 1975; Nevo et al., 1988; Pirlot and Nevo, 1989)?

The results of these previous observations are contradictory. Leder (1975) found clear reductions in the visual system of *Spalax leucodon*; e.g. a very thin optic nerve without myelinated fibers, and a lateral geniculate body, from which only the ventral nucleus was developed. Cooper et al. (1993) could show that not all parts of the visual system are small in *Spalax*: The accessory visual system, consisting of the dorsal and lateral terminal nuclei and the suprachiasmatic nucleus, involved in the maintenance of hormonal and behavioral circadian rhythms, seems to be well developed, whereas the pretectum, tectum and the dorsal nucleus of the lateral geniculate body are extremely reduced in size. The minute size of the dorsal lateral geniculate body was also shown in an electrophysiologically controlled tracing study by Rehkämper et al. (1994).

On the other hand, Bronchti et al. (1989) described the dorsal nucleus of the lateral geniculate body as a prominent structure; studying the 2-deoxyglucose (2-DG) uptake, they claimed that this normally visual thalamic structure receives auditory input in *Spalax*. Cortical regions that correspond to visual areas in sighted rodents were also labelled by 2-DG after acoustic stimulation, indicating a very elaborate auditory system (and hence intermodal compensation). Leder (1975), however, denied a compensatory "over-development" of other brain structures in *Spalax leucodon*, whereas Pirlot and Nevo (1989) mentioned that olfaction and vocalisation may be reinforced as compensation for the loss of vision.

The structures measured in the latter study (e.g. olfactory cortices plus amygdala), however, are functionally heterogeneous.

Recently, NECKER et al. (1992) mapped the body representation in *Spalax* with electrophysiological methods and found that large parts of the occipital cortex correspond to somatosensory, but not to auditory nor to visual stimuli.

The aim of the present investigation is to measure volumes of defined brain regions with well known functions (as, e.g. the auditory nuclei and the motor nucleus of the trigeminal nerve). These volume data will be evaluated using brains from species of Insectivora as the reference base.

The value of Insectivora for such a comparison lies in the fact that a group of insectivore-like species gave rise to all other mammalian orders (THENIUS and HOFER, 1960). Though recent Insectivora species have adapted to a great variety of ecological niches, some species have remained generalists and are thought to most closely resemble the early mammalian condition.

With regard to the brain, species of the subfamily Tenrecinae, the Madagassian hedgehog-like tenrecs, have the smallest brains in relation to their body weights (STEPHAN et al., 1991), i.e. they represent, amongst studied species, the lowest level of encephalization in eutheria and therefore are a useful base to evaluate the development of the brain or brain parts in other mammalian species.

Materials and Methods

Ten *Spalax ehrenbergi* from the Anza-population (chromosomal species 2n = 60) in Israel were investigated. Body weights were determined and brain weights were measured immediately after perfusion.

After paraffin embedding, eight brains were cut in the coronal plane, one brain in the sagittal plane and one in the horizontal plane. Additionally, eight laboratory rat brains (*Rattus norvegicus*, pigmented strain) were cut coronally. The sections were alternately stained for rough endoplasmatic reticulum with cresyl violet, for cell bodies using MERKER's (1983) protocol and for myelinated fibers (GALLYAS, 1971).

Measurements of brain structure volumes were performed using camera lucida drawings of the contours of the brain regions.

The following structures were delineated:

Tegmentum, tectum, cerebellum, diencephalon, telencephalon (Table 1); bulbus olfactorius, paleocortex, amygdala, septum, striatum, schizocortex, hippocampus, neo(=iso-)cortex (Table 2); cerebellar nuclei, ncl. motorius nervi trigemini, superior and inferior colliculi, cochlear nuclei (ncl. cochlearis dorsalis, ncl. cochlearis ventralis), complexus sensorius nervi trigemini (= ncll. pontinus et spinalis nervi V), and funicular nuclei (= dorsal column nuclei: ncl. gracilis, ncl. cuneatus medialis, ncl. cuneatus externus) (Table 3).

For the measurement of the major brain regions, 50 to 60 camera lucida drawings of equidistant sections, covering the whole brain, were used for planimetry on a digitizer tablet. The resulting areal values for a given structure were multiplied with the section thickness and the distance between sections, to arrive at the serial section volume. The sum of the volumes of the major brain regions in serial sections is considerably smaller than the volume of the fresh brain, due to tissue shrinkage during the histological processing. The extent of shrinkage is different in each brain. To obtain comparable values, the figures have to be converted to the fresh brain value by multiplying each structural volume with the conversion factor (C) for shrinkage:

C= volume of fresh brain/sum of serial section volumes

For the measurement of tiny nuclei, the distances between the camera lucida drawings used for the determination of the major brain region volumes were too large and the magnification too low; therefore, other sets of camera lucida drawings with higher magnification and smaller distances between the sections were used for planimetry (ZILLES et al., 1982). The resulting volumes were also corrected, with the shrinkage factor for the whole brain, to fresh volume.

For the evaluation of the calculated volumetric data for brain structures in *Spalax* and *Rattus* allometric methods are used. The relationship between brain or brain structure size and body size has been found to be represented best by the power function

$$y = b \cdot x^a,$$

or, in its logarithmic form,

$$\log y = \log b + a \cdot \log x,$$

in which y is the brain or brain structure size, b the intercept of the allometric line with the abscissa, x the body weight, and a the slope of the line.

The slope of the allometric line can be calculated with different methods (e.g., least-square regression, major axis, reduced major axis). In the present investigation the slopes used were published by STEPHAN et al. (1991), who calculated them with a canonical analysis based on the slopes of allometric lines for 56 mammalian families or subfamilies. The canonical analysis assigns each group its appropriate weight when calculating a common slope, i.e. families well represented in the material and covering a larger range of body weights have a greater influence on the overall slope than those poorly represented or of limited body weight range.

For the calculation of size indices (SIs), the allometric line was placed through the Tenrecinae, a group of terrestrial, hedgehog-like insectivores from Madagascar with the smallest brains in relation to body sizes among eutheria (Fig. 3). All points on this line represent a SI of 100; e.g., indices of 200 mean twice the relative size of brains or brain components of Tenrecinae.

Though volumes of individual specimens of *Spalax* and *Rattus* are given in Tables 1-3, it must be emphasized, that in any case only one allometric size index is calculated for a given structure, using the average body weight and the average structural volume (see Table 4). This procedure must be applied in order to prevent the mixing of intra- and interspecific allometries, which follow different allometric rules (HERRE and RÖHRS, 1990). Usually, intraspecific slopes of allometric lines are much shallower than those obtained from interspecific comparisons.

On the other hand, the calculation of one SI per species prevents the use of statistical methods to show the significance of the differences found. For most structures, however, the size differences between *Spalax* and the rat or tenrecine insectivores are so large, that there is no overlap of the data points and thus a statistical proof is not necessary (see Fig. 3a, neocortex). Only these large differences shall be discussed in detail in the following sections.

Table 1. Body and brain weights, and volumes (in mm^3) of fundamental brain regions in *Spalax* and *Rattus*.

	No	Body (in g)	Brain (in mg)	Tegmentum	Cerebellum	Tectum	Diencephalon	Telencephalon
Spalax								
	293	77.2	1948	281.20	282.76	26.43	109.75	1080.94
	294	141.4	2164	300.54	305.35	26.55	196.74	1148.08
	296	138.3	2141	309.93	311.77	28.87	199.62	1187.59
	377	86.7	2191	309.06	316.27	30.63	199.40	1220.66
	380	87.6	1990	279.73	298.99	27.62	172.45	1061.04
	423	152.0	2028	298.27	281.81	29.76	181.77	1127.96
	424	114.0	1987	288.12	293.78	26.66	182.24	1106.14
	426	105.6	1920	278.39	299.65	26.21	173.33	1043.56
	(n=8)	112.8	2046	293.16	298.80	27.84	187.04	1121.99
	±SD	28.4	104	12.99	12.47	1.70	11.15	61.60
Rattus								
	1793	224.1	1660	252.72	215.51	46.81	139.17	904.57
	1794	219.5	1770	289.74	239.91	45.25	157.37	920.77
	1800	259.3	1820	295.98	243.04	50.97	149.30	954.77
	1801	225.4	1660	266.95	221.61	43.76	148.61	878.45
	1802	259.3	1830	310.75	247.00	53.78	151.28	984.40
	1807	190.1	1680	267.17	210.39	45.12	149.28	905.68
	1814	193.9	1700	268.45	215.17	44.89	148.49	902.39
	1832	298.0	2040	340.88	265.97	55.23	180.95	1097.09
	(n=8)	233.7	1770	286.58	232.33	48.23	153.06	943.52
	±SD	36.4	129	28.95	19.61	4.46	12.32	70.48

Table 2. Volumes (in mm^3) of telencephalic brain structures in *Spalax* and *Rattus*.

No	Bulbus olfact.	Paleocortex	Amygdala	Septum	Striatum	Schizocortex	Hippocampus	Neocortex
Spalax								
293	64.97	95.52	37.64	24.73	135.34	63.08	120.23	539.43
294	79.52	109.37	42.10	28.02	138.10	75.93	128.24	546.79
296	74.16	111.99	43.97	27.58	149.40	82.54	126.15	571.80
377	87.52	117.95	38.38	27.31	144.79	78.46	126.99	599.26
380	65.23	92.81	34.36	22.39	113.43	64.99	127.05	540.79
423	78.36	98.13	37.90	26.82	131.65	90.83	118.83	545.44
424	77.28	98.89	38.85	25.64	129.55	79.99	117.57	538.36
426	69.75	86.70	37.04	22.32	114.79	80.48	108.95	523.51
(n=8)	74.60	101.42	38.78	25.60	132.13	77.04	121.75	550.67
±SD	7.72	10.63	2.99	2.27	12.88	9.14	6.65	23.79
Rattus								
1793	52.64	70.76	34.29	19.51	89.43	63.04	96.56	478.35
1794	43.76	67.36	35.25	17.92	94.10	71.19	98.62	492.56
1800	73.17	81.50	37.14	18.82	87.64	65.71	98.69	492.10
1801	44.67	78.90	32.47	17.59	90.05	52.86	87.75	474.17
1802	55.11	85.84	37.73	21.80	93.55	69.42	96.35	524.59
1807	69.45	83.74	28.15	16.74	82.65	53.51	92.45	478.99
1814	67.52	88.74	32.84	16.56	86.12	52.44	98.65	459.51
1832	91.54	96.98	38.09	21.59	100.26	77.24	110.66	560.72
(n=8)	62.23	81.73	34.50	18.82	90.48	63.18	97.47	495.13
±SD	16.26	9.54	3.34	2.03	5.45	9.43	6.55	32.59

Allometric size indices of various brain structures in insectivores (*Chrysochloris asiatica*, *Chr. stuhlmanni*, *Talpa europaea*, *T. micrura*, *Parascalops breweri*, *Scalopus aquaticus*) from Stephan et al. (1991) are partly included in the tables of this study to allow comparisons with insectivore species that occupy a similar ecological niche, the subterranean environment.

The original research reported herein was performed under the official German regulations for research on animals.

Results

The transversal sections through the *Spalax* brain (Fig. 1) show some of the measured structures and their delineation, which followed the brain atlas of the hedgehog in Stephan et al. (1991). Additional information for the delineations was extracted from Paxinos and Watson (1986).

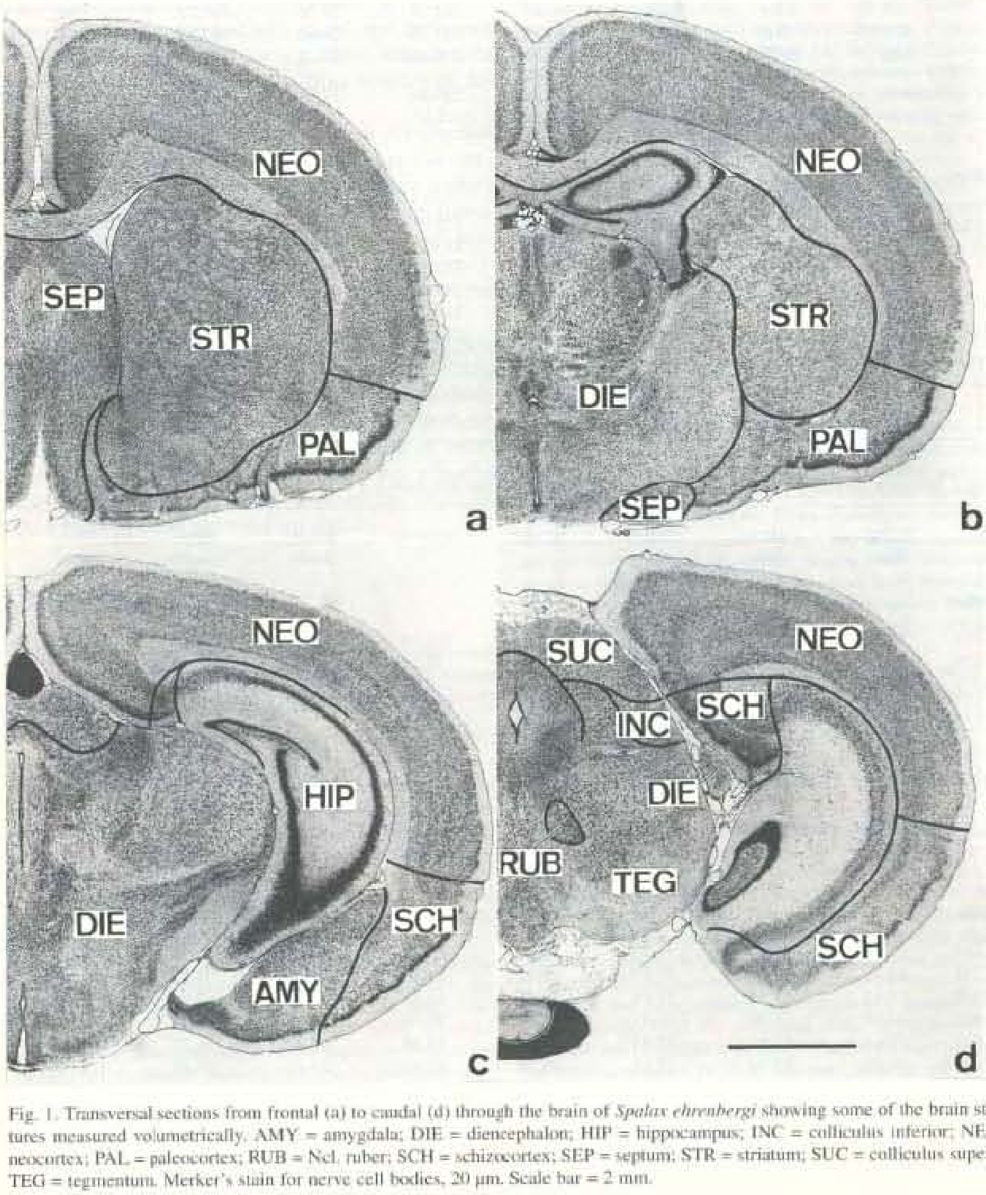

Fig. 1. Transversal sections from frontal (a) to caudal (d) through the brain of *Spalax ehrenbergi* showing some of the brain structures measured volumetrically. AMY = amygdala; DIE = diencephalon; HIP = hippocampus; INC = colliculus inferior; NEO = neocortex; PAL = paleocortex; RUB = Ncl. ruber; SCH = schizocortex; SEP = septum; STR = striatum; SUC = colliculus superior; TEG = tegmentum. Merker's stain for nerve cell bodies, 20 µm. Scale bar = 2 mm.

Data on body- and brain-weights and volumes of brain structures in *Spalax* and the laboratory rat are given in Tables 1-3. It can be seen that *Spalax* is a relatively small animal, reaching only half the size of the rat, but it's brain weight is slightly higher than that of the rat.

The allometric comparison, based on species from the insectivore subfamily of Tenrecinae (*Tenrec ecaudatus*, *Setifer setosus*, *Hemicentetes semispinosus*, and *Echinops telfairi*) reveals that *Spalax* has a rather high encephalization (Size Index SI = 258), that means its brain is 2.6 times larger than in a hypothetical tenrec of the same body weight (Table 4). Among visual structures a respective cortex is missing (NECKER et al., 1992) and the lateral geniculate extremely reduced (REHKÄMPER et al., 1994). Among the brain structures measured in *Spalax* the greatest enlargement is observed in the neocortex (SI = 788; Fig. 3a). A more than 4fold enlargement is found for the striatum (SI = 454), more than three times larger are the diencephalon (SI = 387), the cerebellar nuclei (359), the whole cerebellum (337), and the ncl. motorius nervi V (327; Fig. 3c). Smaller than in an average tenrec are the cochlear nuclei (SI = 68, that is only 2/3 of their size in a tenrec; Fig. 3d), the olfactory structures bulbus olfactorius (75; Fig. 3b) and paleocortex (82), the gracile nucleus (80), and the sensory trigeminal complex (91).

Compared to tenrecs, the rat brain is only 1.4 times larger, so the encephalization of the rat is much lower than that of *Spalax*. Correspondingly, most SIs for the measured brain structures are lower than in *Spalax*, but the sequence of enlargement shows some similarities between both rodent species.

In the laboratory rat the greatest enlargement is, like in *Spalax*, observed for the neocortex, followed by the diencephalon and the striatum. The lowest SIs are found for the olfactory bulb, the paleocortex, and the sensory trigeminal complex.

A comparison between the two rodent species reveals that the allometric size index for brain weight in *Spalax* is nearly double that of the rat (258:138 = 1,87times).

This proportion is approximately the same for some brain structures, as e.g., the neocortex (1,89x), hippocampus (1,89x), the olfactory bulb (1,91x), schizocortex (1,94x), telencephalon (1,95x), and diencephalon (1,96x), whereas other structures show greater dissimilarities. Especially large in comparison to the rat is the ncl. motorius nervi trigemini (3,1x, Fig. 2), followed by cerebellar nuclei (2,6x), striatum (2,4x) and cerebellum (2,1x). Smaller than in the rat are the cochlear nuclei (both, dorsal and ventral cochlear nuclei; 0,63x and 0,76x, respectively) and the inferior colliculi (0,72x).

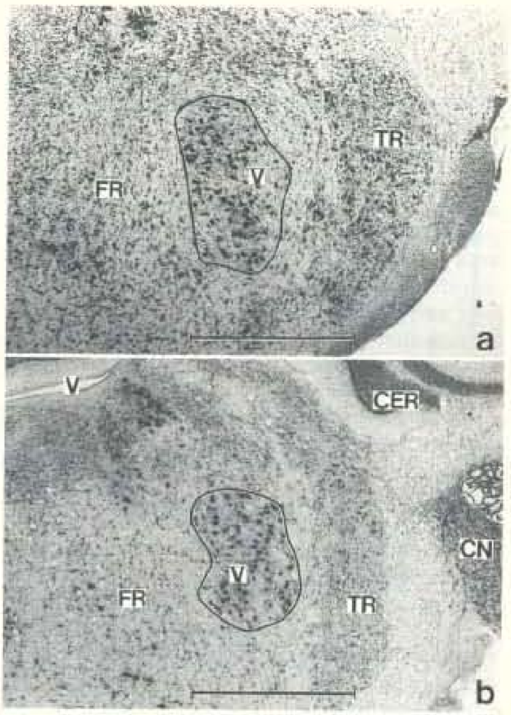

Fig. 2. Transversal sections of the Nucleus motorius nervi trigemini in *Spalax ehrenbergi* (a) and *Rattus norvegicus* (b). CER = cerebellum; CN = cochlear nuclei; FR = Formatio reticularis; TR = Complexus sensorius nervi trigemini; V = ventricle; V = Ncl. motorius nervi trigemini.
Merker's stain for nerve cell bodies. 20 µm. Scale bar = 1 mm.

Discussion

The discussion will follow several lines:

First, a comparison of our encephalization-data in *Spalax* with data reported in the literature a) on *Spalax* and (b) on other rodent species,

second, a discussion of the brain composition in *Spalax* (a) compared to data in the literature on *Spalax*, (b) in comparison to tenrecs, (c) the laboratory rat and (d) subterrestrial insectivores (Chrysochloridae, Talpidae), which, like *Spalax*, are animals with reduced visual capabilities (data taken from STEPHAN et al., 1991),

third, the question of possible compensations for the loss of visual acuity by other sensory modalities will be discussed, and,

fourth, a description of the pecularities of the *Spalax* brain will be given.

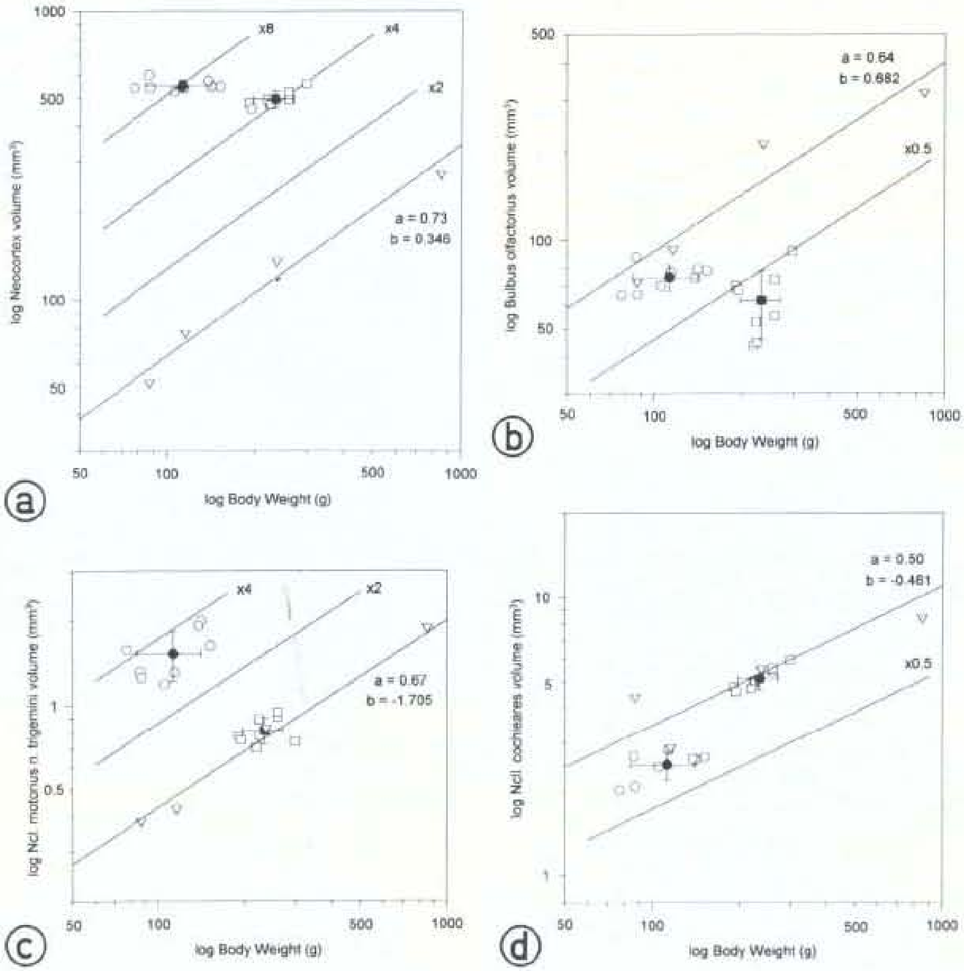

Fig. 3. Volumes of brain structures (in mm³) related to body weight (in grams).
a) Neocortex, to show the greatest structural enlargement in *Spalax* as compared to tenrecs;
b) Bulbus olfactorius, demonstrating reduction of size in *Spalax*; and especially in the rat;
c) Ncl. motorius nervi trigemini and
d) Ncll. cochleares, as an example of unchanged sizes in the rat in comparison to tenrecs, but clear enlargement resp. reduction in *Spalax*.

❑ = *Rattus norvegicus*; ❍ – *Spalax ehrenbergi*; ▽ – Tenrecinae.

The solid line in the diagram represents the allometric line of four species of Madagassian insectivores, the Tenrecinae. Members of this subfamily are the least encephalized mammals, i.e. their brains are the smallest in relation to their body sizes (STEPHAN et al., 1991).

The mode of representation shall give an impression of the variability of brain structure sizes and of body weights in *Spalax* and the rat; it must be emphasized, however, that only one size index is calculated for a given structure, using the average body weight and the average structural volume (see Material and Methods).

1. Encephalization in Spalax in comparison to

– Spalax-data in the literature

Body weights of the four chromosomal species of *Spalax* were published by NEVO et al. (1986). For the 2n=60 species from Anza (which is also the origin of the material studied here) they gave a mean body weight of 124 g (n=304). This is larger than our average of 113 g (SD = ± 28 g), but within the range covered by our material.

In 1988, NEVO et al. also published brain weights from individuals of the Anza population. Their average brain weight of 1.731 g (n=20) is smaller than in our sample and, in combination with the higher body weight (136.2 g), leads to a lower size index (SI = 193). In their study of brain structure size, PIRLOT and NEVO (1989) used only one specimen from the Anza population, a male with a high body weight and a rather large brain, resulting in a SI of 188, i.e. in the same order of magnitude as in the NEVO et al. (1988) study.

Despite these differences in brain- and body weights given for *Spalax* in the various publications, it can be said that in all cases the resulting SIs show a clearly higher encephalization for the mole rat than for the rat.

– other rodent species

PIRLOT and NEVO (1989) also published encephalization indices for seven rodent species, including *Spalax*. The mole rat showed the highest encephalization, clearly higher than e.g., the golden hamster or the mouse.

Besides the data on other rodent species, given by PIRLOT and NEVO (1989), there are additional data of brain and body weights in Rodentia (STEPHAN and DIETERLEN, 1982). Except *Colomys*, all species reported by STEPHAN and DIETERLEN (1982) are more or less terrestrial, with some digging ability, but not to such an extent as *Spalax ehrenbergi*, a true fossorial and subterranean species, which rarely is active above ground (WALKER, 1975).

In Insectivora, STEPHAN et al. (1991) could show that fossorial species have relatively larger brains than terrestrial animals. The same is observed in rodents. *Spalax* is situated above the SI range of terrestrial rodents, which all have the ability to perform some digging, since most of these species spend a considerable amount of time in underground burrows. A SI higher than that of *Spalax* is found in *Colomys goslingi* (SI = 268) another highly specialized rodent species with semiaquatic life habits.

2. Brain composition in Spalax compared to

– Spalax-data in the literature

PIRLOT and NEVO (1989) also studied the brain composition in the four chromosomal species of *Spalax* (one specimen each) and compared it with that of six other rodent species, preferentially living more above the ground. Although the slope of the allometric line and the reference group (basal insectivores) are different from those used in this study, some general trends are the same: The neocortex, diencephalon, striatum and cerebellum are the structures with high SIs in *Spalax*. The olfactory bulb shows a rather small SI, but within the range covered by the other six rodent species. In contrast, PIRLOT and NEVO (1989) state the rhinencephalon (RH = paleocortex + amygdala + schizocortex in their measurements) to be larger in *Spalax* than the average of the six other species, and suggest that the olfactory components may be quantitatively important in species active under ground. Such may be seen particularly in true subterranean blind animals, in which olfaction and vocalization may be reinforced as compensation for the loss of vision (PIRLOT and NEVO, 1989).

We have subdivided this rhinencephalic complex into olfactory paleocortex, amygdala, and schizocortex, and found the SI of the true olfactory structure (PAL) to be nearly as low as that of the olfactory bulb (Table 4). Thus, the higher SI of the rhinencephalon in PIRLOT and NEVO (1989) was apparently a result of a rather large amygdala and schizocortex, which at least do not exclusively serve olfactory functions.

– tenrecs

As mentioned in the results section, the greatest enlargement of any measured brain structure in *Spalax* compared to tenrecs is observed for the neocortex, followed by the striatum, diencephalon, the cerebellar nuclei, the total cerebellum, and the ncl. motorius nervi trigemini.

– the laboratory rat

A similar sequence of structures with the greatest enlargements in comparison to Tenrecinae is found in the rat. In this species the neocortex is the most enlarged brain structure, followed by the diencephalon and the striatum.

It should be kept in mind, however, that, although the sequence of structures is similar, the absolute enlargement is much less in the rat than in *Spalax* for almost all measured structures.

– subterranean insectivores

The brain of *Spalax* is rather large in relation to its body size, resulting in a SI, which is higher than that

Table 3. Volumes of brain structures related to certain functional systems.

No.	TCN	mot V	SUC	INC	CON	DCO	VCO	TR	FUN	FGR	FCM	FCE
Spalax												
293	7.521	1.579	11.69	14.74	2.015	0.605	1.411	12.74	2.695	0.485	1.465	0.745
294	10.981	2.007	13.74	12.81	2.590	1.085	1.506	16.25	3.689	0.303	2.096	1.290
296	10.916	1.925	13.62	15.25	2.612	1.107	1.505	15.93	2.870	0.504	1.636	0.730
377	9.983	1.322	13.66	16.97	2.671	1.147	1.524	11.19	2.654	0.437	1.209	1.008
380	7.866	1.258	15.39	12.23	2.073	0.890	1.183	11.16	2.414	0.372	0.963	1.079
423	8.803	1.636	13.54	16.22	2.653	0.985	1.668	17.88	3.072	0.596	1.290	1.186
424	9.999	1.313	12.96	13.70	2.809	1.226	1.583	17.63	3.241	0.539	1.489	1.213
426	8.811	1.193	11.56	14.65	2.447	0.797	1.650	14.89	3.801	0.428	1.549	1.824
(n=8)	9.359	1.529	13.27	14.57	2.484	0.980	1.504	14.71	3.055	0.458	1.462	1.134
± SD	1.313	0.311	1.23	1.62	0.290	0.206	0.154	2.71	0.497	0.094	0.335	0.347
Rattus												
1793	5.438	0.893	19.13	27.68	4.968	1.891	3.077	15.53	3.116	0.620	1.595	0.901
1794	5.293	0.704	15.97	29.28	4.716	2.193	2.523	19.26	3.629	0.692	1.770	1.167
1800	6.136	0.912	16.53	34.44	5.392	2.497	2.895	15.16	3.717	0.653	1.974	1.090
1801	6.007	0.775	15.75	28.01	5.022	2.411	2.611	14.98	3.269	0.600	1.695	0.974
1802	6.660	0.947	19.79	33.99	5.539	2.576	2.963	17.11	3.779	0.670	1.983	1.126
1807	5.337	0.773	16.42	28.70	4.762	2.096	2.666	13.72	3.166	0.686	1.489	0.991
1814	5.509	0.751	15.80	29.09	4.588	1.773	2.815	13.04	3.118	0.820	1.408	0.890
1832	7.632	0.741	21.13	34.10	5.952	2.536	3.416	19.91	3.940	0.910	1.955	1.075
(n=8)	6.002	0.812	17.57	30.66	5.117	2.247	2.871	16.09	3.467	0.706	1.734	1.027
± SD	0.812	0.091	2.12	2.96	0.470	0.307	0.288	2.48	0.335	0.106	0.226	0.103

Abbreviations of brain structures as in Table 4.

Table 4. Allometric size indices (based on body weight) in *Spalax* and *Rattus* compared to the least encephalized mammals, the tenrecine insectivcres

Structure		a	b	*Spalax*	*Rattus*	Proportion *Spalax*/*Rattus*
Brain weight	ENC	0.66	1.544	258	138	1,87
Tegmentum	TEG	0.61	0.868	222	139	1,60
Cerebellum	CER	0.68	0.552	337	160	2,11
Tectum	TEC	0.58	0.141	130	147	0,88
Diencephalon	DIE	0.65	0.350	387	197	1,96
Telencephalon	TEL	0.68	1.242	258	132	1,95
Bulbus olfactorius	BOL	0.64	0.682	75	39	1,91
Paleocortex	PAL	0.63	0.799	82	42	1,96
Amygdala	AMY	0.60	0.232	133	77	1,74
Septum	SEP	0.61	-0.129	193	91	2,12
Striatum	STR	0.67	0.089	454	191	2,38
Schizocortex	SCH	0.64	0.062	325	167	1,94
Hippocampus	HIP	0.57	0.581	216	114	1,89
Neocortex	NEO	0.73	0.346	788	416	1,89
Cerebellar nuclei	TCN	0.68	-0.979	359	140	2,56
Ncl. motorius n. V	V	0.67	-1.705	327	107	3,06
Colliculus superior	SUC	0.58	-0.287	166	144	1,15
Colliculus inferior	INC	0.58	-0.061	108	149	0,72
Ncl. cochleares	CON	0.50	-0.461	68	97	0,70
Ncl. cochlearis dorsalis	DCO	0.50	-0.867	68	108	0,63
Ncl. cochlearis ventralis	VCO	0.51	-0.700	68	89	0,76
Ncll. pontinus + spinalis n. V	TR	0.59	-0.001	91	65	1,41
Funicular nuclei	FUN	0.62	-0.927	138	100	1,39
Ncl. gracilis	FGR	0.62	-1.515	80	79	1,02
Ncl. cuneatus medialis	FCM	0.62	-1.300	156	118	1,33
Ncl. cuneatus externus	FCE	0.61	-1.400	159	93	1,72

of subterrestrial insectivores, i.e. Chrysochloridae and Talpinae (Table 5).

When averaging the SIs of the major brain structures in the six species of Insectivora (taken from STEPHAN et al., 1991) and comparing these with the SIs in *Spalax*, the greatest deviations of SIs in *Spalax* are seen in the following structures: neocortex (+108%), diencephalon (+85%), cerebellum (+60%), striatum (+56%), and schizocortex (+54%). These deviations are larger than the enlargement of the total brain in *Spalax* (encephalization + 49%).

On the other hand, there are structures, which clearly have lower SIs in *Spalax* than in subterrestrial Insectivora. These are the paleocortex (PAL = -32%), and the olfactory bulb (BOL = -22%). With respect to the size of olfactory structures, there seems to exist a difference between the two subterranean insectivore families: In both *Chrysochloris* species the BOL and PAL are larger than in an average tenrec, whereas in the talpid species these structures are approximately of the same size as in tenrecs, or show a tendency toward reduction. In fact, the European mole has small olfactory structures, which are allometrically almost identical to their sizes in *Spalax*.

Besides the major brain regions, a number of smaller brain components has been measured in a limited number of Insectivora, among which there were two species with subterrestrial life habits, the African golden mole, *Chrysochloris stuhlmanni*, and the European mole, *Talpa europaea* (STEPHAN et al., 1991).

A striking result is the huge enlargement of the ncl. motorius nervi trigemini (activating the chewing muscles) in *Spalax* (SI +170%) as compared to fossorial insectivores, which clearly reflects another

Table 5. Allometric size indices (based on body weight) in *Spalax* in comparison to subterranean Insectivora.

Species/Structure	n	ENC	TEG	CER	TEC	DIE	TEL
Spalax ehrenbergi	8	258	222	337	130	387	258
Chrysochloris asiatica	1	153		144		188	175
Chrysochloris stuhlmanni	2	184		169		209	211
Talpa europaea	2	160		244		205	159
Talpa micrura	2	200		259		242	214
Parascalops breweri	3	181		226		221	205
Scalopus aquaticus	1	163		220		188	177

Species/Structure	BOL	PAL	AMY	SEP	STR	SCH	HIP	NEO
Spalax ehrenbergi	75	82	133	193	454	325	216	788
Chrysochloris asiatica	101	130	114	152	266	150	189	296
Chrysochloris stuhlmanni	118	166	148	159	297	174	188	391
Talpa europaea	75	85	110	138	257	205	185	327
Talpa micrura	102	120	142	185	323	231	212	468
Parascalops breweri	96	115	121	159	341	229	225	432
Scalopus aquaticus	86	111	105	129	258	281	193	357

Species/Structure	V	CON	DCO	VCO	TR	FUN	FGR	FCM	FCE
Spalax ehrenbergi	327	68	68	68	91	138	80	156	159
Chrysochloris stuhlmanni	140	78	90	71	163	136	154	142	109
Talpa europaea	102	65	81	56	99	224	133	320	166

Abbreviations of brain structures as in Table 4.

type of digging executed by *Spalax*. This species uses the teeth for loosening the soil and shows a number of adaptations in this context: rapidly ever-growing, curved incisors, specialized jaws with strong muscles, and a flattened head for bulldozing the loose soil outside the tunnel system (Nevo, 1991).

In contrast to *Spalax*, fossorial insectivores do not use the teeth for digging. In *Talpa*, the greatly enlarged forehand executes lateral stroke movements that demand finely tuned muscular contractions. Unlike talpids, Chrysochloridae burrow by means of their leathery snout and head strokes. The soil is also loosened by the forepaw, which has an elongated third finger with a powerful claw, and thrusts the loosened soil behind the animal's body in a motion similar to running, not in a lateral stroke movement as the mole (Puttick and Jarvis, 1977).

The smallest SIs are found for the cochlear nuclei of *Spalax*. These nuclei comprise the second neuron of the auditory pathway, originating in the cochlea. From the cochlear nuclei in the brain stem, the information is further processed in the lateral lemniscus via the nuclei of the trapezoid body and superior olive to the inferior colliculus, finally reaching the medial geniculate and auditory cortex. The small size of the cochlear nuclei in *Spalax* is in accordance with the measurements of auditory nuclei in *Talpa* and *Chrysochloris*, which also show similar reduced sizes. In the rat, however, the cochlear nuclei were of the same size as those of tenrecs. This points to the fact that in both orders, insectivores and rodents, true subterranean species tend to have smaller cochlear nuclei than species more confined to a life on the ground.

The nuclei of the dorsal columns [gracile nucleus (FGR), medial cuneate (FCM), and external cuneate nuclei (FCE)] receive general somatosensory information from the body, the FGR via the gracile fascicle from the hindlimb region, FCM and FCE via the cuneate fascicle from the forelimb region. The complex of dorsal column nuclei shows slightly enlarged SIs in *Spalax* (1.4 times) compared to tenrecs. This SI value is similar to that of *Chrysochloris*, whereas these nuclei are much larger in *Talpa*. Within the complex, the gracile nucleus of *Spalax* is only four fifth that of an average tenrec, but the nuclei receiving somatosensory information from the upper half of the body are clearly enlarged (up to 1.6 times for the FCM).

The greatest SI for any of these dorsal column nuclei among the subterranean species studied is calculated for the FCM (and, to a smaller degree, for the FCE) in the European mole, which can be explained by the intense use of the forepaws for the spade-like digging movements executed by this species.

3. Are there compensations for the loss of visual acuity?

Subterranean mammals generally display regressive stages of eye development from medium to minute eyes completely covered by skin. Apart from the Chrysochloridae and perhaps *Notoryctes*, which are insufficiently investigated, *Spalax* probably has the most reduced eye (SANYAL et al., 1990; DE JONG et al., 1990; COOPER et al., 1993). By contrast, its other sensory modalities seem to be well developed (NEVO,1990). Speculations about the reasons for the reduction of disused brain structures, such as parts of the visual system in *Spalax*, reach from savings of energy to competition for brain space (DIAMOND, 1996).

Possible modalities for the compensation of visual loss could be: audition, olfaction, the vibrissae system, and/or general somatosensory information.

– Audition

Evoked potential recording from the midbrain revealed the best hearing sensitivity between 0.5 and 1 kHz in *Spalax*. Compared with the rat, the frequency range is two octaves smaller in *Spalax*, but the basilar membrane is 3,3 mm longer (BRUNS et al., 1988), indicating an acoustic fovea for low frequency hearing. It has been observed, however, that time parameters in the vocal repertoire may be more important for *Spalax* than exact frequency information (NEVO et al., 1987; BURDA et al., 1989). BRONCHTI et al. (1989), dealing with the auditory activation of primary visual targets in *Spalax*, found an unusual 2-DG uptake in the dorsal nucleus of the lateral geniculate body. Their most striking finding, however, was a marked 2-DG labelling of cortical regions that correspond to visual areas in sighted rodents.

It is surprising that this presumed enlargement of the auditory system on the level of the diencephalon and telencephalon is not reflected in the size of auditory nuclei in the brain stem, as e.g. the cochlear complex with its subdivisions (Table 3). These nuclei are of the same size in tenrecine insectivores and the laboratory rat, but in *Spalax* they only reach two thirds of this size.

HEFFNER and HEFFNER (1992a) studying the hearing and sound localization in *Spalax*, found that these mammals are remarklably insensitive to sound, have an unusually poor high-frequency hearing, and even their low-frequency hearing is not exceptional, but the only aspect of their audiogram that is not degenerate. Therefore, the subterranean habitat is associated with a dramatic reduction in hearing abilities in *Spalax*.

Similar observations were made in a hearing study in the naked mole rat, *Heterocephalus glaber* (HEFFNER and HEFFNER, 1993). The major auditory nuclei in the brain stem of *Heterocephalus* are present and of normal configuration, but seem to be relatively small. Our measurements in *Spalax* are in accordance with these statements by HEFFNER and HEFFNER (1993).

They support the theory of HEFFNER and HEFFNER (1992b) according to which the ability for sound localization is correlated with stereopsis and small foveae or areae centrales since the biological role of sound localization is to help to focus on an object of interest with both eyes. This is not given in the blind mole rat.

– Olfaction

The olfactory bulb and the paleocortex of *Spalax* show SIs approximately two times larger than those of the rat. Whether this enlargement can be interpreted as a compensatory effect seems doubtful. Although rodents are macrosmatic, they do not reach the size of olfactory structures observed e.g. in terrestrial insectivores.

Measurements in six other rodent species (PIRLOT and NEVO, 1989) resulted in an allometric SI for the olfactory bulb of 60 as opposed to 75 in *Spalax* (reference group: basal insectivora = 100). Among the six rodent species in their comparison, three had larger, and three smaller olfactory bulbs than *Spalax*. These figures indicate that olfactory structures are of average size in *Spalax* compared to other rodent species and there is no sign of a compensatory overdevelopment of this system in the mole rat.

– Sensory trigeminal and general somatosensory systems

Measurements of the sensory trigeminal complex (= ncl. principalis nervi trigemini and ncl. spinalis nervi trigemini) in *Spalax* show that this complex nearly reaches the SI of an average tenrec. In comparison to the rat, the sensory trigeminal complex is 1.4 times enlarged in *Spalax*, as opposed to a 1.9fold enlargement of brain size; i.e., the sensory trigeminal complex does not seem to be disproportionally enlarged in *Spalax*.

The size of the dorsal column nuclei, i.e. the gracile and cuneate nuclei, mediating mechanoreceptive (extero- and proprioceptive) information from the postcranial body has already been discussed in relation to their sizes in subterrestrial insectivores. Compared to the rat, a slight enlargement of the complex is observed, which is mainly due to clearly larger cuneate nuclei in *Spalax*, whereas the allometric size of the gracile nucleus is nearly identical to that of the rat.

It should be mentioned, however, that the size of dorsal column nuclei may not be the best measure for statements about the importance of the somatosensory system, because of the following reason: The heavily myelinated fibers of neurons mediating mechanoreception in the spinal ganglia reach the spinal cord via the dorsal root. There they divide into an ascending branch running in the dorsal funiculus to the dorsal column nuclei, the other branch of the primary afferent neuron ends on neurons in the ncl. proprius of the spinal cord. From here, mechanoreceptive information reaches the lateral ventroposterior nucleus of the thalamus via the ventral spinothalamic tract (Tracey, 1985). This shows that mechanoreception is not exclusively mediated to the brain via the dorsal funiculus and dorsal column nuclei, but also in the lateral funiculus. If this second pathway is especially elaborated in *Spalax* is unknown.

Measurements of the thalamic relais nuclei of the somatosensory system reveal a 1,4 times larger VPM/VPL complex in *Spalax* than in the rat (Mann et al., 1997), e.g. its enlargement is identical to that of the sensory trigeminal nuclei.

The electrophysiological investigation by Necker et al. (1992) showed that *Spalax* has a well developed somatosensory cortex. The cortical map includes a large representation of the vibrissae system, and a hindlimb area, that is much smaller than that of the forelimb. Estimations of the areal size of the somatosensory representation on the dorsal surface of the cortex show a larger area in *Spalax* than in the rat, supporting the hypothesis of a caudally expanded somatosensory cortex.

In fact, volume measurements of the somatosensory cortex show that this structure is 1,7 times larger in *Spalax* compared to the rat (Mann et al., 1997). It may well be that *Spalax* has increased the neuronal potential of the isocortex in order to intensify the analysis of somatosensory information. This might primarily affect the proprioceptive data which are correlated with muscular activity during the special mode of digging.

4. *Peculiarities of the Spalax brain*

Our comparative approach revealed that *Spalax* has a relatively large brain with some brain regions contributing much to the overall enlargement. All investigators of the *Spalax* brain agree, that the (main) visual system shows a pronounced reduction in size, but the often postulated over-development of the auditory and olfactory systems as a compensation for the loss in visual acuity can hardly be found.

Compared to the rat, the cochlear nuclei are even smaller in *Spalax* and both, the sensory trigeminal and dorsal column nuclei slightly enlarged (but to a lesser degree than the enlargement of total brain size), whereas the olfactory structures olfactory bulb and paleocortex are two times larger in *Spalax*. They do not reach, however, the size of these structures in tenrecs.

If none of these sensory structures shows an enlargement in *Spalax* that surpasses that of the brain size, other brain parts must be clearly enlarged to account for the overall larger brain of the mole rat. These are the neocortex, striatum, diencephalon, and cerebellum among the major brain regions. Measurements in a great variety of primates species show that an enlargement of the neocortex is almost always accompanied by a size increase of the diencephalon and the striatum, e.g., structures, which are closely interconnected with the neocortex (Stephan et al., 1988).

Often, large brains are associated with complex social behaviour and communicative interactions, remarkable learning and enlarged integrative capacities (Stephan et al., 1988; Rehkämper and Zilles, 1991). *Spalax*, however, is an animal that does not live in structured social groups, instead, individuals are very aggressive toward each other (Nevo, 1991). Since the morphological substrate for communicative interactions and integrative capacities is mainly located in the nonprimary areas of the neocortex (Passingham and Ettlinger, 1974) and a detailed analysis of nonprimary neocortical areas in the brain of *Spalax* is not available, a definite statement on this issue is impossible.

On the other hand, structures which show a great enlargement in the *Spalax* brain, such as the cerebellum, the cerebellar nuclei, striatum, and ncl. motorius nervi trigemini have one function in common: They are all completely or partly engaged in motor processing.

In this context, it is worth mentioning that the enlargement of the total neocortex in *Spalax* (1.9 times larger than in the rat) is distinctly surpassed by that of the motor cortex, which is 3,1times larger (REHKÄMPER et al., 1994) and thus seems to be the main contributor to the overall enlargement of the neocortex in the blind mole rat.

The ncl. motorius nervi trigemini is reached by cortical afferents via the cortico-nuclear tract and its target comprises the chewing muscles. These muscles seem to be very large in *Spalax* and enable the animal to dig with its incisors. The brain of *Spalax* seems to be highly adapted to the motorical demands of its peculiar digging behaviour and the associated somatosensory processing.

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